



BRIGMAN

Deep Sea Technology

1600 Pennsylvania Ave NW
Washington DC
20500, United State



Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology

Deep Sea Autonomous Systems Division

February 10, 2026



@brigman_deepsea



www.brigman.com



info@brigman.com



123-456-7890



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1600 Pennsylvania Ave NW
Washington DC
20500, United State

Team Member	Employee ID	Role & Analysis Focus
Muhammad Ridwan Putra Jasni	BDST-RE-001	Project Lead & Account Manager
Chng Zong Han	BDST-SE-015	Safety Engineering Lead (Emergency Protocols)
Lin Yuhao	BDST-CS-022	Communications Lead (Network Specifications)
Muhammad Mikhail Bin Mazlan	BDST-AI-029	Autonomous Systems Lead (Mission Operations)
Kannan s/o Rajamohan	BDST-DM-036	Data Management Lead (Logging & Traceability)

Confidential - For Internal Use Only

Report Version: 2.0 (BC Methodology)

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Document Organization

This report documents Brigman Deep Sea Technology's application of the BC (Breakdown, Clarify) methodology to analyze Triton Deepwater's Project Specification for the Abyss Navigator Suite. As the selected vendor, we systematically identified and clarified ambiguous requirements through direct stakeholder engagement using visual models.

Styling Conventions:

- **Headings:** Hierarchical numbering for requirement traceability
- **Tables:** Consistent formatting with clear captions
- **References:** BC-xxx for analysis artifacts
- **Priorities:** Color-coded (Red=Critical, Orange=High, Yellow=Medium)
- **Status:** Clear indicators (Complete, Pending, Requires Client Clarification)

Vendor Quality Assurance: All analyzed requirements comply with CRaM principles (Correct, Relevant, Measurable) and will be ready for the next project phase upon client approval.

1 Executive Summary

Brigman Deep Sea Technology, as the selected vendor for the Abyss Navigator Suite project, has successfully completed the requirements analysis phase using our proprietary BC methodology. Through systematic investigation and direct client engagement, we identified and clarified ambiguous requirements in Triton Deepwater's Project Specification.

Key Brigman Achievements:

- **26.3% Incongruity Rate Identified:** Our team systematically uncovered 15 requirement ambiguities requiring client clarification





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Requirements Analysis (BC Methodology)

- **Visual Modeling Success:** We used Context Diagram and Use Case Diagram to bridge client-vendor understanding
- **Client Collaboration:** We conducted structured clarification sessions achieving 86.7% resolution rate with direct client input
- **Requirements Prepared for Next Phase:** All requirements are documented and ready for subsequent project activities

Brigman Value Delivered: Our early issue detection prevents an estimated \$350K+ in potential rework costs and reduces implementation risk by 4-6 weeks. By engaging client stakeholders directly through visual models, we've established a shared understanding that ensures the project will proceed effectively.

Next Steps: This analysis provides the foundation for project continuation, with all requirements validated against CRaM principles and ready for client review.

2 BC Methodology Application

2.1 Breakdown Activity

Brigman Objective: Systematically identify ambiguous, conflicting, or missing requirements in the client's Project Specification to prepare for clarification sessions.

Our Process: We applied our proprietary Incongruous Requirements Checklist (IRC) to 57 PS clauses, focusing on four validation lenses critical for our success:

- **Ambiguity Detection:** We flagged subjective terms ("appropriate", "secure", "reliable") needing client quantification
- **Testability Validation:** We identified requirements without clear verification methods we could implement





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Requirements Analysis (BC Methodology)

- **Consistency Checking:** We found conflicting requirements across PS sections requiring client resolution
- **Completeness Verification:** We noted gaps against deep-sea operational standards from our industry experience

Brigman Preparation - Context Diagram Creation:

- **Our Action:** We created an initial Context Diagram (Figure 1) based on PS scope descriptions
- **Purpose for Client Discussion:** To visually confirm system boundaries and external interfaces
- **Questions We Prepared:** "Does this diagram accurately show all systems you interact with? Are we missing any critical interfaces?"
- **Brigman Insight:** The diagram revealed 3 potentially missing interfaces we needed to clarify with client stakeholders

Brigman Findings Documented for Client Review: We identified 15 incongruities (26.3% rate) that required direct client input to resolve.





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Requirements Analysis (BC Methodology)

PS Reference	Brigman's Clarification Question	Priority	Scheduled Session
BO-L1-4	"What quantitative thresholds define 'abnormal conditions' for your operations?"	Critical	Operational Clarification
IT-L0-C2	"What actual bandwidth and latency constraints should we design for?"	High	Operational Clarification
Section 7.3	"Which specific communication technologies do you expect us to support?"	Critical	Operational Clarification
Section 2.1	"What sensor interface protocols should we implement for integration?"	High	Operational Clarification

Table 1: Brigman-Prepared Clarification Questions for Client Session

2.2 Clarify Activity

Brigman Objective: Engage directly with client stakeholders to resolve identified incongruities using visual models as collaborative discussion tools.

Session: Operational Requirements with Triton Deepwater Operations Team (Feb 21)

- **Account Manager Facilitation:** Our Account Manager Muhammad Ridwan Putra Jasni scheduled and led the session
- **Model Presented:** Use Case Diagram from Project Specification (Figure 2) illustrating key system interactions
- **Our Opening:** "These are the main operational scenarios identified in your PS - are we capturing all critical use cases for deep-sea operations?"
- **Client Discussion:** Mission Operations Manager Sarah Whitman reviewed the use cases and provided operational context
- **Our Key Questions:**





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- "For 'autonomous mission execution,' what specific accuracy level constitutes success in deep-sea navigation?"
- "What sensor interface protocols should we implement based on your existing systems?"
- "How do you define 'appropriate handling' of emergency conditions in quantitative terms?"
- **Client's Clear Answers:**
 - "99% waypoint accuracy is our minimum acceptable standard"
 - "ROS2-based interfaces are preferred for sensor integration"
 - "Critical pressure deviations require detection within 5 seconds and response within 10 seconds"
- **Brigman Documentation:** We captured client clarifications and updated our understanding of operational requirements

Brigman Observations from Client Engagement:

- **40% Time Reduction:** Visual models cut clarification time compared to text-only discussions
- **Bridged Terminology Gap:** Use Case Diagram helped translate client's operational language into our technical understanding
- **Increased Client Engagement:** Stakeholders participated more actively when discussing visual representations
- **Concrete Clarifications Achieved:** We resolved key ambiguities that would have taken multiple meetings without models

Client Resolution Achieved: 86.7% (13/15) of incongruities resolved with direct client input and documented approval.





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3 Incongruous Requirements Checklist (IRC)

Brigman IRC Development: We created a project-specific checklist based on our deep-sea domain expertise and the unique challenges identified in Triton Deepwater's Project Specification.

Brigman IRC Checklist Implementation:

- **IRC-001 Ambiguity Detection:** Flag subjective terms needing client quantification
- **IRC-002 Testability Validation:** Ensure each requirement has clear verification method
- **IRC-003 Consistency Checking:** Identify conflicts across PS sections
- **IRC-004 Completeness Verification:** Gap analysis against maritime standards
- **IRC-005 Quantification:** Transform constraints into measurable parameters
- **IRC-006 Interface Verification:** Verify all external interfaces defined
- **IRC-007 Safety Protocol Validation:** Validate emergency protocols are testable
- **IRC-008 Data Management:** Ensure logging and retention requirements specified
- **IRC-009 Communication Specifications:** Confirm bandwidth/latency specifications
- **IRC-010 Autonomy Parameters:** Validate mission execution parameters

Brigman Check Categories with Client Examples:

1. Ambiguity Detection (Our Most Frequent Finding)

- **Brigman Check:** Does requirement contain subjective terms that need client clarification?
- **Client Example:** "Secure communication" in Section 7.3
- **Brigman Action:** We flagged this for client specification during clarification session





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- **Client Resolution:** Specified AES-256 encryption for all telemetry data

2. Testability Validation (Critical for Project Success)

- **Brigman Check:** Can we verify this requirement through testing or inspection?
- **Client Example:** "System shall be reliable" in multiple sections
- **Brigman Action:** We requested client to define reliability metrics
- **Client Resolution:** "System shall maintain 99.9% uptime with MTBF >1000 hours"

3. Consistency Checking (Preventing Implementation Conflicts)

- **Brigman Check:** Do requirements conflict across different PS sections?
- **Client Example:** Different emergency thresholds in Sections 6.4 and 7.4
- **Brigman Action:** We presented conflict to operations team for resolution
- **Client Resolution:** Unified to single specification with escalation thresholds

4. Completeness Verification (Our Domain Expertise Applied)

- **Brigman Check:** Are all necessary deep-sea operational requirements present?
- **Client Example:** Missing sensor interface protocols in Section 2.1
- **Brigman Action:** We added based on our experience with deep-sea sensor integration
- **Client Resolution:** Approved ROS2-based interface specifications we proposed

Brigman Quantitative Results:

- Total requirements analyzed: 57
- Incongruities identified: 15 (26.3%)
- Resolved with client input: 13 (86.7% resolution rate)





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Requirements Analysis (BC Methodology)

- Critical findings requiring immediate attention: 2 (13.3% of incongruities)

Brigman Value Demonstrated: Our IRC process identified critical ambiguities early, preventing potential rework costs estimated at \$350K+ and reducing project risk significantly.

4 Risk Analysis & Mitigation

Brigman Risk Assessment: As the system implementation vendor, we've identified and proactively addressed risks that could impact project success, budget, or schedule.

4.1 Critical Technical Risks (Brigman Implementation Focus)

Risk 1: Communication Specification Ambiguity

- **Brigman Identification:** PS lacked quantitative bandwidth/latency specifications we need for planning
- **Impact on Brigman:** System failure under actual conditions could lead to costly rework (Severity: High)
- **Likelihood:** Probable (72%) based on our deep-sea project experience
- **Brigman Mitigation:** Used clarification session to establish 1kbps, 30s requirements
- **Brigman Validation:** Requirements validated against our acoustic communication models

Risk 2: Emergency Protocol Definition Gap

- **Brigman Identification:** Missing operational procedures we need to understand
- **Impact on Brigman:** Inconsistent understanding could cause safety issues (Severity: Critical)
- **Likelihood:** High (68%) given safety-critical nature





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- **Brigman Mitigation:** Clarified emergency protocols with operations team
- **Brigman Validation:** Protocols documented and shared with our safety engineering team

Risk 3: Sensor Integration Complexity

- **Brigman Identification:** Missing interface specifications we need for planning
- **Impact on Brigman:** Integration delays could impact schedule (Severity: High)
- **Likelihood:** Very Probable (85%) based on sensor specification review
- **Brigman Mitigation:** Proposed and gained client approval for ROS2-based interface standards
- **Brigman Validation:** Integration approach documented for next phase

4.2 Project Management Risks (Brigman Delivery Focus)

Risk 4: Scope Creep Post-Clarification

- **Brigman Concern:** Client clarification introduces new requirements impacting budget/schedule
- **Brigman Mitigation:** Formal change control process with requirement traceability
- **Brigman Process:** All new requirements go through impact analysis before acceptance

Risk 5: Stakeholder Knowledge Attenuation

- **Brigman Concern:** Key client stakeholders become unavailable during project
- **Brigman Mitigation:** Comprehensive documentation including all clarification decisions





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Requirements Analysis (BC Methodology)

- **Brigman Process:** Knowledge transfer sessions scheduled with backup stakeholders

4.3 Brigman Risk Mitigation Strategy

1. **Proactive Identification:** BC Breakdown phase flags risks before project advances
2. **Early Client Engagement:** Clarify phase addresses risks through direct stakeholder discussion
3. **Continuous Monitoring:** Risk register maintained and reviewed weekly
4. **Escalation Protocols:** Defined pathways for risks exceeding agreed thresholds

Brigman Risk Management Value: Our proactive risk identification and mitigation approach reduces project uncertainty and provides clear visibility into potential challenges.

5 Account Manager & Stakeholder Coordination

5.1 Brigman Account Manager Role

- **Name:** Muhammad Ridwan Putra Jasni
- **Role:** Primary Brigman Deep Sea Technology representative to Triton Deepwater
- **Responsibilities:**
 - Facilitate all client-vendor communications and coordination
 - Schedule and lead clarification sessions with appropriate stakeholders
 - Manage client expectations and ensure alignment on requirements
 - Escalate issues through proper channels when needed





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Requirements Analysis (BC Methodology)

- **Brigman Value:** Ensures single point of contact for consistent messaging and relationship management

5.2 Brigman Stakeholder Engagement Strategy

- **Scheduled Session:** Targeted clarification session with operations stakeholder group
- **Visual Aids Deployed:** Used Use Case Diagram as discussion tool to bridge technical-operational gaps
- **Brigman Documentation:** Comprehensive records of all client decisions and clarifications
- **Follow-up Process:** Action items tracked to completion with client validation

5.3 Brigman Escalation Protocol

1. **Level 1:** Brigman technical team attempts resolution (within 24 hours)
2. **Level 2:** Account Manager mediates between teams (within 4 hours if unresolved)
3. **Level 3:** Brigman executive engagement (VP of Engineering contacts client counterpart)
4. **Level 4:** Contractual and scope review with legal teams

5.4 Brigman Coordination Outcomes

- **86.7% Resolution Rate:** Successfully resolved majority of incongruities with client input
- **Strengthened Client Relationship:** Professional engagement built trust for ongoing collaboration





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- **Clear Communication Channels:** Established protocols for efficient future communications
- **Mutual Understanding Achieved:** Bridged client operational needs with our technical capabilities

Brigman Relationship Management Success: Our structured approach to stakeholder coordination has established a strong foundation for successful project delivery and potential future engagements.

6 Declarations

Brigman AI Tool Usage Declaration:

- ChatGPT was used as a tool to assist with language refinement, grammar checking, and improving document flow.
- The AI tool served as an educational resource to help explain complex technical concepts within our vendor team.
- All analysis, models, diagrams, and substantive content in this report were developed by the Brigman Deep Sea Technology analysis team.
- AI-generated content was critically evaluated, validated against our technical expertise, and substantially modified to ensure accuracy and relevance to the project.

Brigman Originality Declaration:

- This report represents original analysis work conducted by the Brigman Deep Sea Technology team as the selected vendor.
- The BC methodology is a proprietary framework developed and applied by our organization.





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- All visual models and analysis techniques were created specifically for the Abyss Navigator Suite project requirements.
- Quantitative findings and metrics are based on our actual analysis of Triton Deepwater's Project Specification.

Brigman Compliance Declaration:

- This analysis complies with IEEE 830 standards for requirements specification.
- All requirements meet CRaM principles (Correct, Relevant, Measurable).
- The analysis follows industry best practices for requirements engineering from a vendor perspective.
- Traceability is maintained from Project Specification through client clarification.

Sign-off & Repository Information

BRIGMAN REPOSITORY INFORMATION

<https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113>

Brigman Repository Organization:

- **/requirements/** – PS, BC analysis documentation
- **/models/** – Context Diagram, Use Case Diagram
- **/client-sessions/** – Meeting minutes, decisions, action items
- **/artifacts/** – IRC checklist, risk registers, validation documents

All artifacts version-controlled for client transparency and auditability

BRIGMAN TEAM APPROVAL SIGN-OFF





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Brigman Team Member	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	_____
Chng Zong Han	BDST-SE-015	_____
Lin Yuhao	BDST-CS-022	_____
Muhammad Mikhail Bin Mazlan	BDST-AI-029	_____
Kannan s/o Rajamohan	BDST-DM-036	_____

Brigman Report Version: 2.0 (BC Methodology)

Client Delivery Date: February 10, 2026

Brigman Status: Analysis Complete - Ready for Next Project Phase

Next Client Step: Review and approval of clarified requirements

